RUMBLE OS • CLINICAL HYDROTHERAPY

Poolside Laminated Reference • Sheet 1: Foundational Decompression & Multi-Planar Gait

SHEET 1 OF 5 • FOUNDATIONS

Cervical Fusion (C4–C7):	Neutral spine strictly locked; no head thrusting.	**Right Shoulder:**	Below 80° abduction.	**Left Knee:**	Max 45° flexion.	**Water:**	32°C–34°C ideal.

h1 Buoyancy Spinal Decompression

25 MIN • RESTORATIVE DECOMPRESSION

Target: Lumbar disc & facet offload, C4–C7 axial decompression

Equipment: Deep water noodle, cervical flotation collar

Mechanism: Eliminates 90% gravity compression

1. Buoyant Vertical Traction — 6m 15s

Place noodle under armpits in chest/deep water so feet hang completely suspended. Relax abdominal wall and let legs dangle. **Zero foot contact.** Breathe deeply into diaphragm.

2. Gentle Water Knee-to-Chest — 6m 15s

While vertically suspended, slowly draw both knees towards chest. Hold 3s to open posterior facet joints. Lower with control. **3 sets x 10 smooth reps.**

3. Aquatic Torso Pendulum — 6m 15s

Holding noodle securely, let legs and pelvis sway smoothly side to side like a gentle pendulum. Keep head and cervical spine completely still and aligned.

4. Supine Water Float & Reset — 6m 15s

Transition onto back with cervical pillow and knee noodle bolster. Complete passive floating surrender. Slow 4s inhale, 6s exhale to extinguish autonomic pain loops.

h2 Aquatic Multi-Planar Gait & Walking

20 MIN • ADAPTIVE MOBILITY & GAIT RETRAINING

Target: Symmetrical stride length, gluteus medius, hip flexor activation

Equipment: Chest-deep pool lane, pool rail

Focus: Upright posture, no forward trunk flexion

1. Forward Water Marching — 5m 00s

Chest-deep water. Drive knee upward smoothly (max 70° hip flexion) with conscious heel-to-toe rolling push-off. Maintain upright chest and natural reciprocal arm swing submerged.

2. Backward Heel-to-Toe Walking — 5m 00s

Step backward toe-to-heel. Feel the drag of water fire hamstrings and glutes without lumbar hyperextension. Maintain level pelvis throughout stride.

3. Lateral Sidestepping — 5m 00s

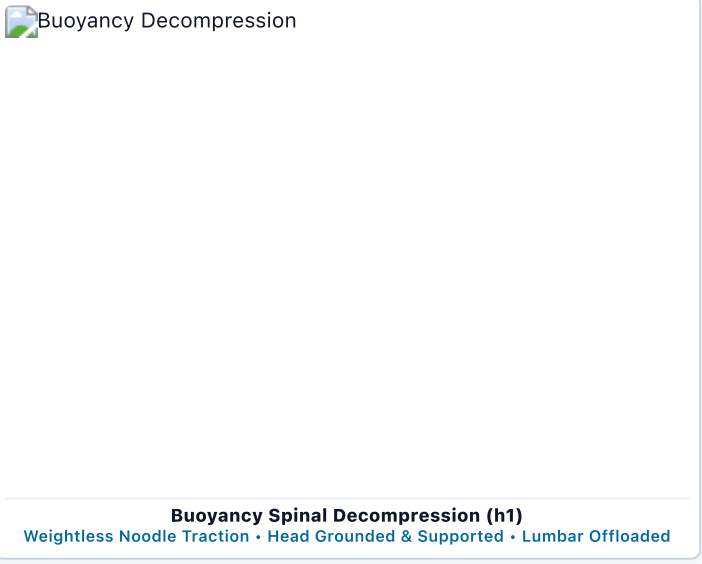
Face pool edge. Step sideways across lane, pushing against water column with outer hip. Step wide, close slowly. **Fires gluteus medius to stabilize pelvis & SI joint.**

4. Warm Water Calming Wall Rest — 5m 00s

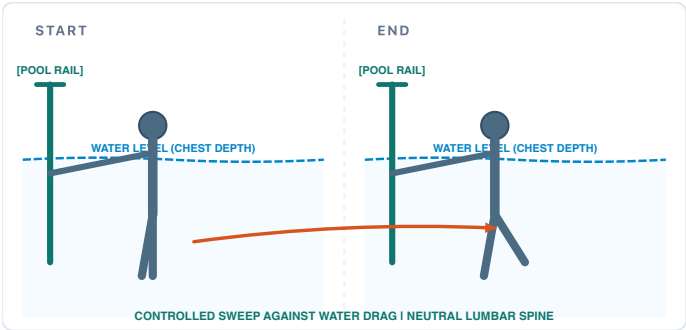
Stand supported with back against pool wall. Hands on rail. Feel hydrostatic pressure support venous return. Gentle deep breathing before exiting.

HIGH-DEFINITION CLINICAL FORM INFOGRAPHICS

Start & Finish Positions • Chest-Deep Fluid Mechanics



Buoyancy Spinal Decompression (h1)
Weightless Noodle Traction • Head Grounded & Supported • Lumbar Offloaded



Aquatic Forward Marching & Multi-Planar Gait (h2)
Chest-Deep High Knees • Submerged Reciprocal Arm Swing • Upright Torso

CLINICAL NOTE: Hydrotherapy allows high muscular recruitment without land gravity impact. If sharp radicular nerve pain or dizziness occurs, exit pool immediately. Keep right shoulder below 80° plane.

h3 Aquatic Hip & Core Stabilization

20 MIN • TARGETED STRENGTHENING

1. Aquatic Standing Hip Abduction — 5m 00s

Stand holding pool rail for balance. Keep torso tall and sweep leg laterally out against water resistance. 3 sets x 12 reps per leg. Controlled tempo; avoid pelvic hiking.

2. Kickboard Core Press-Down — 5m 00s

Hold kickboard flat with both hands at chest level. Engage deep transversus abdominis and push board straight down into water column. 3 sets x 10 reps (3s hold at bottom).

3. Aquatic Bicycle Pedaling — 5m 00s

Lean back resting upper back on noodle across armpits. Smoothly pedal legs through water column like riding a bicycle. Maintain rhythmic cadence without knee shear.

4. Floating Spine Rest & Elongation — 5m 00s

Supported supine float with noodle under knees. Lengthen spine, relax pelvic floor and deep hip rotators.

h4 Warm Water Relaxation & Lymphatic Flush

20 MIN • LYMPHATIC & AUTONOMIC FLUSH

1. Deep Water Submersion Breathing — 5m 00s

Submerged to clavicles. Inhale 4s through nose; let water hydrostatic pressure assist full, effortless 6s exhalation. Lowers systemic vascular resistance.

2. Aquatic Arm Sweeps (Lymphatic Flow) — 5m 00s

Slow figure-eight arm sweeps submerged below chest line. Elbows soft, hands relaxed. Drives axillary lymphatic drainage without shoulder impingement.

3. Ankle & Foot Water Mobility — 5m 00s

Rest hands on rail at pool steps. Point, flex, and rotate ankles through full fluid range. Activates calf venous muscle pump to clear peripheral fluid.

4. Supported Supine Water Float — 5m 00s

Complete stillness on water surface with neck pillow and knee noodle. Deep parasympathetic reset to downregulate dorsal horn pain gating.

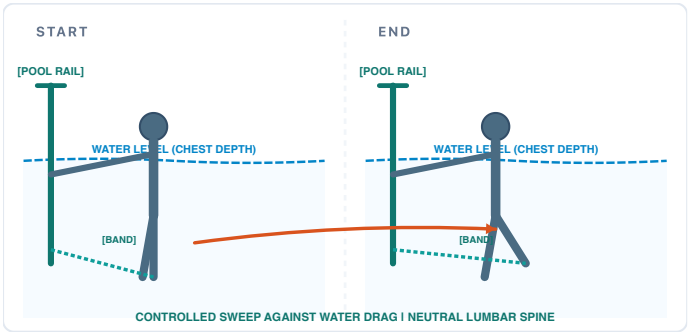
SHORT & EXPRESS AQUA RESETS (A1–A6 LIBRARY)

For quick mid-day decompression

CODE	ROUTINE NAME	TIME	PROTOCOL CIRCUIT SEQUENCE
A1	**Water Reset**	6 min	Forward/backward walking (2m) • Rail knee lifts (1m) • Torso pendulum (1m) • Supported rest (2m)
A2	**Side-Step Reset**	6 min	Lateral side walking (2m) • Wall knee drives (1m) • Hip abductions (1m) • Edge relaxation (2m)
A3	**Balanced Aqua**	14 min	Water walking (3m) • Supported kicks (2m) • Rail abductions (2m) • Low arm sweeps (2m) • Float (3m)
A4	**Low-Impact Aqua**	14 min	Forward/back strides (3m) • Lateral walk (2m) • Leg pendulums (2m) • Chest presses (2m) • Rest (3m)
A5	**Full Aqua Session**	25 min	Water walking (5m) • Cycling motion (3m) • Wall flutter kicks (3m) • Side raises (3m) • Float (5m)
A6	**Aqua Variety**	25 min	Multi-speed walking (5m) • Side strides (3m) • Knee lifts (3m) • Kicks (3m) • Arm scull (3m) • Float (5m)

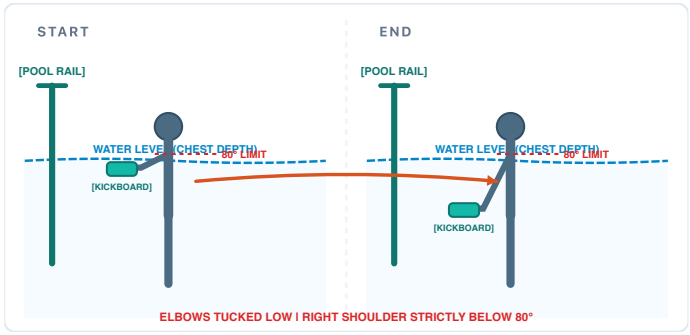
HIGH-DEFINITION CLINICAL FORM INFOGRAPHICS

Pelvic Stability • Kickboard Fluid Resistance



Standing Hip Abduction & Extension (h3)

Rail-Anchored Sweep • Upright Neutral Spine • Gluteus Medius Isolation



Kickboard Core Press-Down (h3)

Bilateral Downward Press • Transversus Bracing • No Cervical Strain

Theraband Setup: Anchor securely to stainless pool rail with slip-knot. **Speed:** 2s pull / 3s return against water drag. **C4–C7:** Cervical chin tuck.

H-MED1 Dynamic Aqua Balance & Banded Core

30 MIN • MEDIUM RESISTANCE

Equipment: Medium Theraband (Green), Pool Rail, Noodle **Focus:** Pelvic stability, anti-shear core bracing, gait power

Phase 1: Dynamic Aquatic Warm-Up — 5m 00s

Forward walking (heel-to-toe) in chest-deep water (2m) • Backward walking engaging hamstrings/glutes (2m) • Lateral side-stepping holding pool rail (1m).

Phase 2: Banded Hip Abduction & Extension — 8m 00s

Anchor band to rail; loop to ankle. Stand tall holding rail. **Abduction:** sweep leg out against band/water tension (3x12/leg). **Extension:** sweep straight back into glutes without arching lumbar spine (3x12/leg).

Phase 3: Pool Wall Pelvic Neutral & Wall Press — 7m 00s

Back flat against wall. Flatten lower spine into wall by bracing deep core (hold 10s x 6 reps). Submerge noodle at chest level; press downward engaging lats without flexing neck (3x10 reps).

Phase 4: Noodle Deep Water Core Suspensions — 6m 00s

Noodle under axillae in deep water. Alternate slow knee lifts toward chest against water resistance (3 sets x 15 reps). Keep torso completely vertical; no swinging.

Phase 5: Buoyancy Decompression Cool-Down — 4m 00s

Supported supine float with noodle under knees and neck collar. Parasympathetic reset: diaphragmatic breathing, feeling warm water drain tension from spine.

H-MED2 Total Body Aquatic Conditioning & Scapular Rowing

45 MIN • MEDIUM RESISTANCE

Equipment: Heavy Theraband, Pool Rail, Kickboard, Water Dumbbells **Focus:** Upper back scapular stabilization, shallow squats without knee shear

Phase 1: Hydrodynamic Gait & Scapular Prep — 10m 00s

Forward, backward, and lateral water strides (5m). Submerged arm sweeps and aquatic chin retractions in chest-deep water (5m). Activates deep cervical flexors safely.

Phase 2: Submerged Rail Band Rows & W-Rows — 10m 00s

Anchor heavy band to rail at waist height. **Low Row:** Pull handles back, squeezing shoulder blades together (elbows strictly below shoulder) (3x12). **W-Row:** Flare hands slightly, squeeze lower traps (3x10).

Phase 3: Banded Pool-Rail Shallow Squat Presses — 10m 00s

Heavy band around rail and pelvis. Facing rail, sink into shallow water squat (max 45° knee bend). Drive up forcefully through heels and glutes against band pull. **4 sets x 10 reps.**

Phase 4: Kickboard Aquatic Core Presses & Twists — 10m 00s

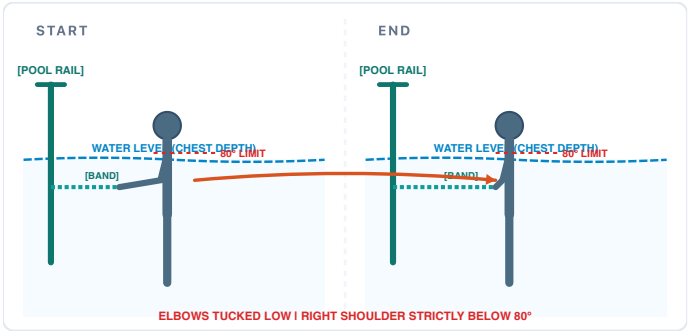
Submerge kickboard vertically with both hands; press down into water column (3x12). Gentle torso rotations with board submerged (eyes strictly forward, cervical locked) (3x10).

Phase 5: Decompression & Ankle Flushes — 5m 00s

Hold rail at pool step. Perform slow ankle circles, calf stretches on pool step edge, and passive buoyant decompression float.

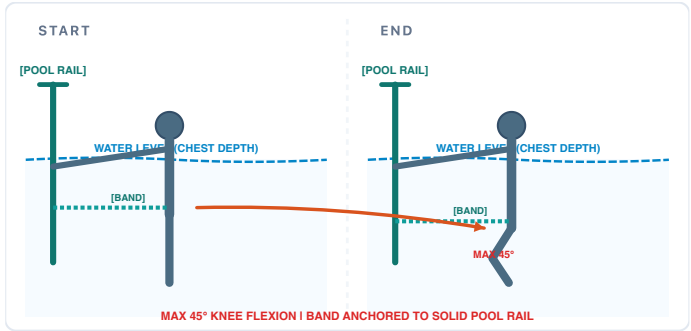
HIGH-DEFINITION CLINICAL FORM INFOGRAPHICS

Rail Band Rowing • 45° Shallow Squats



Submerged Band Low Rows & W-Rows (H-MED2)

Waist-Height Rail Anchor • Elbows Tucked Low • 80° Shoulder Limit Enforced



Rail-Supported Banded Shallow Squats (H-MED2)

Heavy Pelvic Band Pull • Max 45° Knee Bend • Forceful Glute Drive

Equipment: Silver/Black Theraband, Heavy Ankle Loop, Hydro Dumbbells, Flotation Belt. **Core Rule:** Power comes from glutes & core, never lumbar spine.

H-EXP1 Heavy Hydro Resistance & Power Protocol

60 MIN • EXPERT / HIGH RESISTANCE

1. Dynamic Thermal & Gait Warm-Up — 10m 00s

High-knee marching (3m) • Multi-planar lateral walk-outs holding rail (3m) • Submerged rotator cuff pendulum sweeps (4m).

2. Banded Rail Power Squat Pulses — 10m 00s

Heavy Theraband around waist anchored to rail. Perform shallow explosive pulses (30°–45° knee flexion), driving forcefully through glutes against band tension. **4 sets x 12 reps (60s rest).**

3. Submerged Unilateral Rows & Press — 10m 00s

Anchor band to rail. **Left arm:** Heavy powerful rows against band & water drag (4x12). **Right arm:** Controlled micro-range pulls below 80° shoulder elevation (4x8).

4. Banded Knee Extensions & Kicks — 12m 00s

Heavy band on ankle anchored to rail. Facing away: extend knee against band (4x12/leg). Facing side: heavy hip abductions against band (4x12/leg). Builds quad strength without knee shear.

5. Deep Water Core Crucible — 10m 00s

Flotation belt on, deep water. Alternate 45s high-cadence bicycle pedaling with 45s double knee-to-chest tucks. **5 continuous cycles.**

6. Thermal Decompression & Stretch — 8m 00s

Buoyancy-assisted hamstring stretches, pool wall calf decompressions, passive traction hanging vertically from noodle.

H-EXP2 Multi-Planar Hydro Endurance & Kinetic Chain Overhaul

75 MIN • EXPERT / HIGH ENDURANCE

1. Hydrodynamic Activation Circuit — 12m 00s

400m water stride combination: forward walking, backward walking, and carioca crossover steps holding rail for stability.

2. Banded Lateral & Monster Walks — 15m 00s

Heavy loop band around ankles in chest-deep water. Side-step across pool against band/drag (4x15 steps L/R). Monster walks (wide zig-zag strides forward/back) (4x12 strides).

3. Heavy Dumbbell Upper Body Matrix — 13m 00s

Water dumbbells submerged at mid-chest. Submerged bicep curls to low-punches (elbows low) (3x12). Torso turns with dumbbells at ribs (thoracic only, neck locked) (3x12).

4. Explosive Glute Kickbacks with Band — 12m 00s

Band anchored to rail, looped to ankle. Drive straight back into terminal glute extension. Hold 2s peak contraction. **4 sets x 12 reps per leg.**

5. Deep Water Interval Treading — 13m 00s

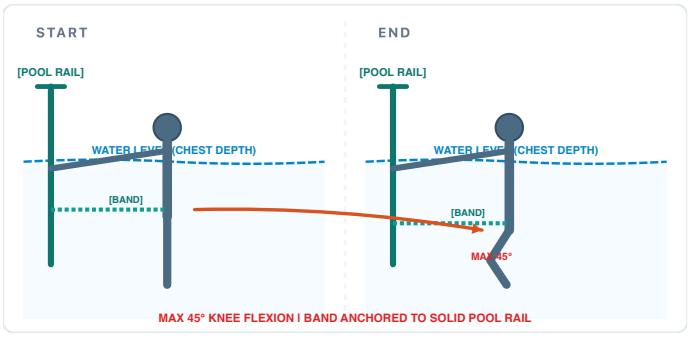
Flotation belt, deep water. 1-minute maximum effort water treading / high-knee churn, followed by 1-minute slow recovery sculling. **6 rounds.**

6. Buoyancy Elongation & Lymphatic Flush — 10m 00s

Full supine float with noodles under head and knees. Passive hydrostatic lymphatic return and complete nervous system downregulation.

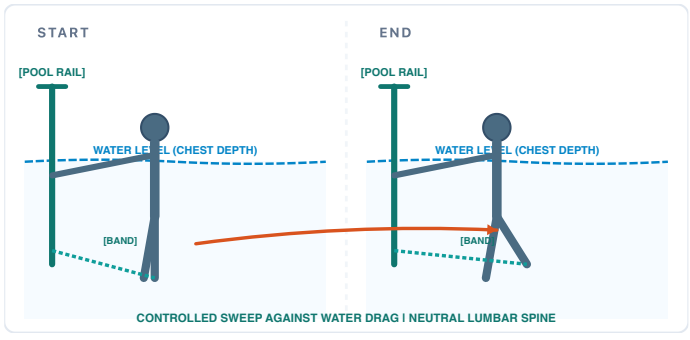
HIGH-DEFINITION CLINICAL FORM INFOGRAPHICS

Power Squat Pulses • Monster Walks • Deep Water Treading



Power Squat Pulses (H-EXP1)

Heavy Rail Band Pull • 45° Explosive Glute Extension • Upright Spinal Neutrality



Banded Monster Walks & Stride Overhaul (H-EXP2)

Heavy Ankle Loop • Wide Multi-Planar Strides • Gluteus Medius & Pelvic Bracing

H-EXP3 Master Hydro Athlete Rehabilitation Protocol

90 MIN • EXPERT / MAXIMUM RESISTANCE

Equipment: Full Hydro Resistance Kit (Heavy bands, Hydro paddles, Flotation belt, Noodles, Rail)

Target: Complete athletic performance overhaul bypassing all land gravity limitations

Phase 1: Multi-Speed Gait & Thermal Prep — 15m 00s

Progressive speed water walking (slow → moderate → fast athletic stride). Submerged shoulder sculling and scapular retractions. Active ankle dorsiflexion against water drag.

Phase 2: Heavy Band Upper Kinetic Chain — 20m 00s

Rail-anchored Low Rows (3x12) • Heavy W-Rows (3x10) • External/Internal shoulder rotation complexes with band (3x10/arm, keeping right arm low) • Scapular protraction/retraction holds (3x12).

Phase 3: Lower Kinetic Chain & Pelvic Power — 20m 00s

Heavy banded rail squats with 3s isometric pause at 45° (4x10) • Banded submerged knee extensions for quadriceps hypertrophy (4x12/leg) • Banded lateral walk-outs and glute kickbacks (4x12/leg).

Phase 4: Deep Water Aquatic Core Crucible — 18m 00s

Suspended scissor kicks against water resistance (4x45s) • Submerged kickboard presses into water column (4x12) • Deep water high-knees interval sprints (flotation belt on): 5 rounds of 1 min on / 1 min off.

Phase 5: Neuromuscular Gait Balance — 10m 00s

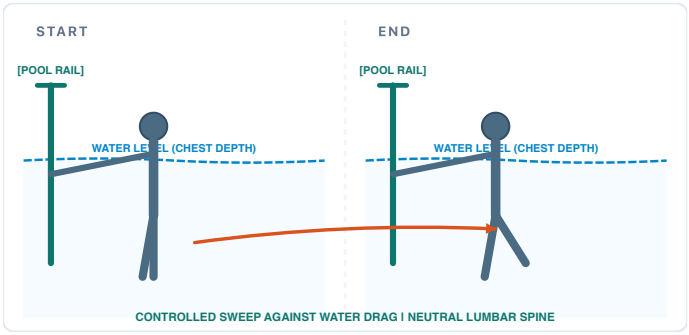
Single-leg balance holds in chest-deep water (holding rail as needed) (2m/leg) • Heel-to-toe tightrope water walking with contralateral arm sweeps (6m). Retrains proprioception without fall injury risk.

Phase 6: Passive Decompression & Surrender — 7m 00s

Noodle-supported vertical spinal traction. Supine float with deep diaphragmatic parasympathetic breathwork. Complete systemic downregulation and mental integration.

MASTER ATHLETE KINETIC CHAIN INFOGRAPHICS

Upper Body Rowing & Lower Kinetic Power



Phase 2: Heavy Band Upper Body Chain

Heavy Rail Rows • Scapular W-Rows • Right Shoulder Protected (<80°)

Phase 3 Lower Power

Phase 3: Lower Kinetic & Pelvic Power

Banded 45° Rail Squats • Submerged Knee Extensions • Glute Drives

POOLSIDE SAFETY RULES & BIOMECHANICAL RESTRICTIONS

1. Cervical Neutrality

C4–C7 anterior fusion hardware requires zero cervical flexion or extension during exertion. Never thrust chin forward against water resistance. Keep gaze locked forward on the horizon.

2. Shoulder Protection

Right shoulder exercises must remain below 80° abduction/flexion. Zero overhead presses or wide flyes. Keep elbows tucked low near ribcage during all pulling and pushing movements.

3. Knee Joint Care

Left knee meniscal preservation: max 45° knee flexion during squats and marches. Avoid sudden pivoting or twisting on a planted foot. Smooth fluid transitions only.

EMERGENCY STOP & EXIT CRITERIA: Stop session immediately and safely exit pool if you experience: (1) Sudden electric or shooting pain down either arm or leg; (2) Dizziness, lightheadedness, or overheating from pool warmth; (3) Chest tightness or shortness of breath; (4) Numbness in hands or feet. Hydrate with electrolytes post-pool.

MANDATORY MEDICAL DISCLAIMER: This hydrotherapy compilation is synthesized for personal physical therapy execution and decision support under clinical guidance. It is not an independent medical diagnosis. Consult your neurosurgeon and physiotherapist before advancing resistance levels.